



D6 Gene expression and homeostasis (HL)

D · Continuity and change · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Not all genes in a cell's genome are expressed at all times; gene expression is regulated by mechanisms such as transcription factors, which control whether RNA polymerase can bind and transcribe a particular gene, and epigenetic modifications (e.g. DNA methylation, histone acetylation), which alter chromatin accessibility and gene expression without changing the underlying DNA sequence. This regulation underlies cell differentiation, allowing cells with identical genomes to develop very different structures and functions, and also underlies homeostatic control, where internal or external signals (such as hormones binding receptors) switch specific genes on or off to help maintain stable internal conditions.

Worked example: follow the reasoning

Explaining hormonal regulation of gene expression to help restore blood glucose homeostasis. Step 1: when blood glucose falls, the hormone cortisol (a steroid hormone) is released and, being lipid-soluble, diffuses through the plasma membrane of target liver cells. Step 2: cortisol binds an intracellular receptor protein, forming a hormone-receptor complex. Step 3: this complex enters the nucleus and acts as a transcription factor, binding to a specific regulatory DNA sequence and promoting transcription of genes encoding enzymes for gluconeogenesis (glucose synthesis from non-carbohydrate sources). Step 4: increased levels of these enzymes raise the rate of glucose production in the liver, releasing more glucose into the blood. Conclusion: by switching on specific genes, the hormone signal directly changes protein levels in target cells, helping to restore blood glucose to its homeostatic set point.

Quick check · multiple choice

Epigenetic regulation of gene expression, such as DNA methylation, typically works by:

- A Changing the base sequence of the DNA
- B Altering chromatin structure/accessibility to transcription machinery without changing the DNA base sequence
- C Directly degrading mRNA molecules in the cytoplasm
- D Adding amino acids to a growing polypeptide chain

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A liver cell and a muscle cell in the same individual contain identical genomic DNA sequences, but express very different sets of proteins. [8]
 - (a) Explain how these two cell types can have identical DNA but different structures and functions. [3]
 - (b) Distinguish, in terms of effect on DNA, between a gene mutation and an epigenetic change such as DNA methylation. [2]
 - (c) Using a named example, explain how a hormone can regulate gene expression in a target cell to help maintain homeostasis. [3]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) Although both cells contain the same genome, differential gene expression means that different combinations of genes are transcribed and translated in each cell type; this is controlled by different transcription factors being active, and by different epigenetic modifications (e.g. patterns of DNA methylation or histone modification) affecting chromatin accessibility in each cell type, so each cell produces a distinct set of proteins suited to its specific structure and function.

(b) A gene mutation changes the actual base sequence of the DNA (e.g. a substitution, insertion or deletion of nucleotides), potentially altering the protein product; an epigenetic change, such as DNA methylation, does not alter the base sequence at all, but instead affects how accessible the DNA is to transcription machinery, changing whether/how much a gene is expressed, and can potentially be reversed without any change to the genetic code itself.

(c) Example: cortisol is released when blood glucose is low; being a lipid-soluble steroid hormone, it diffuses through the plasma membrane of liver cells and binds an intracellular receptor; the hormone-receptor complex acts as a transcription factor, binding to DNA and promoting transcription of genes encoding gluconeogenesis enzymes; this increases glucose production and release into the blood, helping to restore blood glucose concentration to its homeostatic set point.

Quick check answer

B. Epigenetic marks such as methylation change how accessible a gene is to transcription factors and RNA polymerase, altering expression, without altering the underlying genetic code.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.